

Uniform Resource Locator

1. Domain Intelligence

- Domain registration page
- Registration expiry
- Registrar reputation
- DNSSEC (Domain Name System Security Extensions) enabled: security feature of the Domain Name System (DNS) that uses cryptographic signatures, data origin authentication, and data integrity protection
- Historical ownership changes
- ASN (Autonomous System Number) reputation: helps different networks talk to each other
- Hosting provider

2. SSL/TLS Intelligence

- Certificate issuer
- Certificate age
- Certificate lifetime
- TLS (Transport Layer Security) version: security level protocol provides encryption, data integrity, etc
- OCSP (Online Certificate Status Protocol) status: certificate is valid or not
- Certificate transparency logs

3. HTTP Behaviour

- Redirect chain length
- Redirect destinations
- HTTP status code
- Security headers (HSTS, CSP, X-frame)
- Cookie
- Response timing

4. Browser Rendering (*Selenium*)

- Final rendered DOM
- Dynamic JS execution
- Popups
- Permission request
- Downloads triggered
- Clipboard access attempts
- Geolocation requests
- Notification requests
- Camera/microphone requests
- Page screenshots

5. HTML Features

- Number of forms
- Password fields
- Hidden fields

- Iframes
- External resources
- Inline javascript
- Obfuscated scripts
- Meta refresh
- Canvas usage
- Local storage/session storage usage

6. JavaScript Behaviour

- Dynamic code generation
- Encoded or packed scripts
- Network requests
- Fingerprinting
- Event listener
- DOM manipulation
- Keylogging indicators

7. Form Analysis

- Form action domain
- Cross-domain submission
- Presence of password fields
- Credit card fields
- OTP fields
- Hidden redirects

8. Visual Analysis

- Detect login pages
- Compare branding similarity
- OCR the page
- Identify impersonation attempts

9. NPL Features

- Urgency language (“Verify now”, “Account suspended”)
- Credential requests
- Payment requests
- Security-related wording
- Readability metrics
- Language consistency

10. Reputation Intelligence

- Google safe browsing
- Virustotal
- Phishtank
- URLhaus
- Openphish
- AbuseIPDB

11. Network Intelligence

- IP geolocation
- ASN
- Cloud provider
- CDN usage
- Open ports
- Hosting history

1. Domain Intelligence

This tells you **who owns the domain**, **how old it is**, and **where it came from**.

A. Domain registration date (Domain age)

Checks when the domain was first registered.

Example:

Domain:
secure-paypal-login.example-phish.com

Registered:
10 days ago



A brand-new domain is **more suspicious** than one registered years ago.

Compare:

paypal.com
Registered: 1998



vs

secure-paypal-login.example-phish.com
Registered: 10 days ago



Older domains can still be malicious, but very new domains are statistically overrepresented in phishing campaigns.

B. Registration expiry

How long the owner paid for the domain.

Example:

Registered:
2026-07-20



Expires:
2027-07-20

Only 1 year.

Many phishing domains are registered for only one year because attackers don't expect to keep them long.

Legitimate businesses often register for multiple years.

C. Registrar reputation

The company that registered the domain.

Example:

Registrar:
NameCheap



or

GoDaddy
Cloudflare Registrar
Google Domains



This **does not mean the registrar is malicious**. Reputable registrars also host phishing domains because anyone can register one.

Instead, reputation systems may flag registrars that historically see unusually high abuse rates.

D. DNSSEC enabled

DNSSEC proves DNS records haven't been tampered with.

Example:

paypal.com



DNSSEC:
Enabled

fake-paypal-login.com



DNSSEC:
Disabled

Not having DNSSEC is **common**, so it isn't strong evidence by itself.

E. Historical ownership changes

Checks if ownership has changed.

Example:

2019
Owner:
ABC Company



2026
Owner:
Unknown

A domain that suddenly changes ownership and immediately starts hosting login pages deserves closer inspection.

F. ASN reputation

ASN = Autonomous System Number.

Every hosting network has an ASN.

Example:

Cloudflare
ASN13335



or

DigitalOcean
ASN14061



Some ASNs have much higher abuse rates than others, but even major cloud providers host malicious sites, so this is only one feature.

G. Hosting provider

Example:

Hosted by:



AWS
Azure
Cloudflare
DigitalOcean
Hetzner
OVH

Useful for context, not proof of phishing.

2. SSL/TLS Intelligence

HTTPS does **not** mean a site is safe.

Certificate issuer

Example:

Issued by:

Let's Encrypt



or

DigiCert



Millions of phishing sites use Let's Encrypt because certificates are free and easy to obtain.

Certificate age

Example:

Issued:

Yesterday



A certificate created yesterday for a brand-new domain is more suspicious than one that's been active for years.

Certificate lifetime

Example:

Valid:

July → October



or

July → next July



Short validity isn't inherently suspicious because certificate lifetimes are limited by industry rules, but it provides context.

TLS version

Example:

TLS 1.3



Modern.

TLS 1.0



Old and less secure.

OCSP status

Checks whether the certificate has been revoked.

Example:

Status:
Good



or

Revoked



A revoked certificate is a strong warning.

3. HTTP Behaviour

Redirect chain length

Example:

URL entered



↓

site1.com

↓

site2.com

↓

login.php

Multiple redirects are common, but unusually long chains may be suspicious.

Redirect destinations

Example:

google.com



↓

accounts.google.com

Normal.

vs

paypal.com



↓

random-domain.xyz/login

Unexpected.

HTTP status code

Examples:

200



Page loads.

301



Permanent redirect.

302



Temporary redirect.

404



Not found.

500



Server error.

Security headers

Examples:

HSTS
CSP
X-Frame-Options



These headers improve security.

Their absence doesn't prove phishing, but their presence is a positive signal.

Cookies

Check whether cookies are:

Secure
HttpOnly
SameSite



Missing protections can increase risk.

Response timing

Example:

40 ms



vs

3500 ms



Large delays may indicate complex redirect chains or overloaded infrastructure, though network conditions also affect timing.

4. Browser Rendering (Selenium)

Instead of downloading HTML only...

Open Chrome automatically.

Like a human.

Final rendered DOM

JavaScript may completely rewrite the page.

Raw HTML:

Loading...



Rendered page:

Fake PayPal Login



Popups

Example:

```
"Your session expired."
```



Permission requests

Example:

```
Allow notifications?
```



```
Allow location?
```



These requests can be abused for spam or social engineering.

Downloads triggered

Example:

```
download.exe
```



immediately starts downloading.

5. HTML Features

Number of forms

Example:

```
12 forms
```



VS

```
1 login form
```



Password fields

```
<input type="password">
```



Obfuscated scripts

Instead of readable code:

```
login()
```



you see:

```
eval(atob("..."))
```



Common in malware, though some commercial software also obfuscates code.

Meta refresh

Example:

```
<meta http-equiv="refresh">
```



Automatically redirects users.

Network requests

Observe connections made after the page loads.

Example:

```
POST credentials
```



```
↓
```

```
attacker.com
```

Fingerprinting

Collects:

```
Screen
```



```
Fonts
```

```
GPU
```

```
Timezone
```

```
CPU
```

Used for tracking or fraud detection.

Event listeners

Example:

```
keyup  
click  
submit
```



Expected on interactive pages.

DOM manipulation

JavaScript changes page content after loading.

Keylogging indicators

Example:

```
Every keystroke
```



```
↓
```

```
sent to server
```

This is a strong malicious signal if observed.

7. Form Analysis

Form action domain

Example:

Displayed page:

```
paypal.com
```



Form submits to:

```
evil.com
```



A classic phishing pattern.

Cross-domain submission

Form posts data to a different domain than the one shown.

9. NLP Features

Analyze the page's language.

Urgency

Examples:

Verify now!



Act immediately!

Your account will be suspended!

Credential requests

Enter password



Enter OTP

Confirm PIN

Payment requests

Pay ₹500



Security wording

Secure Login



Verified

Encrypted

Attackers often overuse reassuring language.

Readability

Measure how simple or complex the text is.

Language consistency

Example:

English page



↓

Random Russian sentence

or inconsistent grammar.

10. Reputation Intelligence

Ask external services whether they have already seen the URL or domain.

Example:

VirusTotal



↓

Detected by

18 vendors

or

0 vendors



Services include:

- Google Safe Browsing
- VirusTotal
- PhishTank
- URLhaus
- OpenPhish
- AbuseIPDB (for IP reputation)

These provide **independent threat intelligence**, but a brand-new phishing site may not appear in any of them yet.

11. Network Intelligence

IP geolocation

Example:

Hosted:

Germany



This is descriptive, not inherently suspicious.

ASN

Cloudflare

ASN13335



Useful for reputation and infrastructure context.

Cloud provider

AWS

Azure

DigitalOcean



CDN usage

Cloudflare CDN



CDNs are widely used by both legitimate and malicious sites.

Open ports

Example:

80

443

22



Unexpected exposed services can add context during analysis.

How all of these work together

Imagine your analyzer extracts these values:

Feature	Value
Domain age	8 days
Certificate age	2 days
Redirects	5
Password form	Yes
Form submits to another domain	Yes
JavaScript uses <code>eval()</code>	Yes
Notification request	Yes
Brand similarity	97% to PayPal
Google Safe Browsing	Not listed
VirusTotal	0 detections
Screenshot	Fake PayPal page

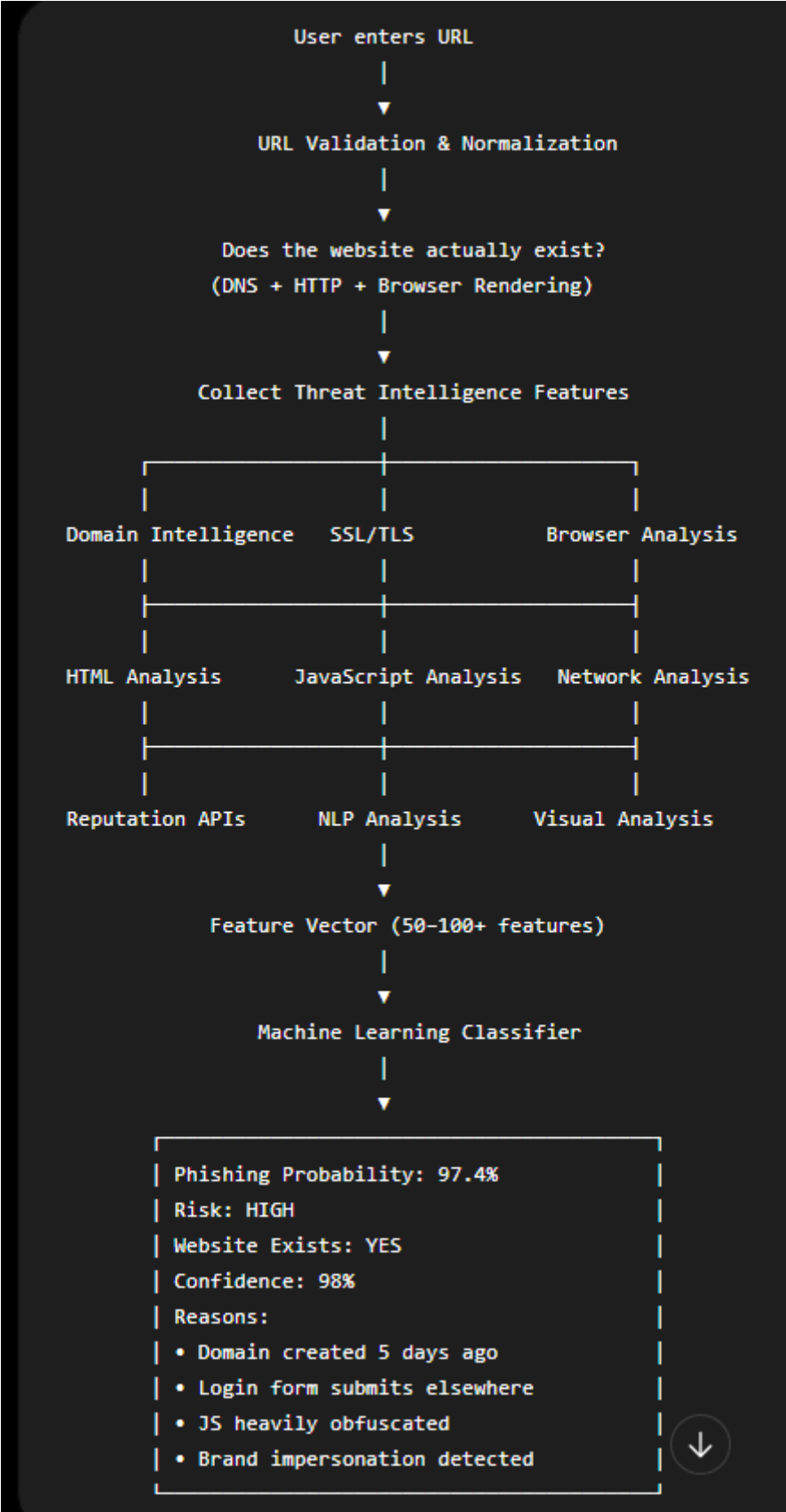
None of these alone proves phishing. However, together they paint a much stronger picture. A machine learning model can learn that this combination of signals is highly associated with phishing and might output:

Phishing probability: 98.6%

Risk: High



Workflow:



Step-1: User enters URL

Example: <https://paypal-login-secure.xyz>

Step-2: Check whether the website exists

- ☐ Is the domain registered?
- ☐ Does DNS resolve?
- ☐ Does it have an IP address?
- ☐ Does it respond over HTTP/HTTPS?
- ☐ Can Selenium load it?

Outcome-1:

Exists: Yes

Outcome-2:

Exists: No

Reason: Domain not registered

Outcome-3:

Exists: Server unreachable

Step-3: Collect evidence

Example:

Domain age: 8 days

Registrar: NameCheap

DNSSEC: Disabled

Certificate: Let's Encrypt

Certificate age: 2 days

Redirects: 4

HTTP Status: 200

Password form: Yes

Notification request: Yes

Screenshot: Captured

VirusTotal: 3 vendors flagged

Google Safe Browsing: Unsafe

Step-4: Convert anything into features

Example:

Feature	Value
Domain age	8
Certificate age	2
Redirect count	4
Password field	1
Notification request	1
Uses iframe	1
DNSSEC enabled	0
Brand similarity	95
VirusTotal detections	3
Safe Browsing	1

Step-5: ML Classifier

Example:

RandomForest or XGBoost

Output:

Phishing probability: 97.8%

Step-6: Risk Calculation

Example:

Probability	Risk
0-20%	Safe
20-40%	Low
40-60%	Medium
60-80%	High
80-100%	Critical

Step-7: Explain the Decision

Example:

Reasons

- ✓ Domain registered only 5 days ago
- ✓ Password form detected
- ✓ Form submits to another domain
- ✓ JavaScript heavily obfuscated
- ✓ Requests notification permission
- ✓ Google Safe Browsing flagged
- ✓ VirusTotal detected by 12 engines
- ✓ Brand similarity with PayPal: 96%

Phase 1 (Foundation)

- FastAPI backend
- URL validation
- DNS lookup
- HTTP request
- Website existence check

Phase 2 (Core Intelligence)

- Domain/WHOIS/RDAP
- SSL/TLS analysis
- HTTP headers
- Redirect analysis

Phase 3 (Dynamic Analysis)

- Playwright browser automation
- HTML parsing
- Form analysis
- JavaScript behavior

Phase 4 (Advanced Intelligence)

- Reputation services
- OCR and screenshot analysis
- NLP on page text
- Network intelligence

Phase 5 (Machine Learning)

- Dataset generation
- Feature engineering
- Model training
- Prediction API
- Explanation of important features